

Olist SQL Analysis DSA Capstone Project

Data Overview

The Brazilian Olist e-commerce dataset contains information about customers, orders, payments, products, sellers, reviews, and geographic data from an online marketplace in Brazil. The analysis used the relevant MySQL tables needed for the eight business questions, including customers, orders, order payments, order items, products, and product category translation. The order data covers September 2016 through October 2018, although the first and last periods contain only partial months. Data quality notes include multiple payment rows for some orders, one order without a matching payment record, and blank product-category values that were handled explicitly in the category analyses.

Business Insight 1 — Customer Spending

Customer spending is concentrated in a smaller high-value group. Query 6 identified 4,296 customers in the High spending tier, with average spending of 929.94 and total spending of 3,995,018.66. The Medium tier was much larger, with 47,889 customers generating 9,166,716.44. This means the business has a relatively small group of higher-value customers while the largest contribution comes from the much larger medium-spending group.

Business Insight 2 — Leading Product Category

Health and beauty was the leading product category. Query 8 found that the category generated 1,258,681.34 in product sales value, representing 9.26% of the total category sales value of 13,591,643.70. Query 4 also placed the same at the top of the category results. This means health and beauty is an important category for the marketplace and deserves attention when reviewing product demand, inventory, and category performance.

Business Insight 3 — Customer Repeat Purchasing

The available data shows no repeat buyers. Query 7 classified all 99,441 customers as one-time buyers, with 0 customers classified as repeat buyers. This means that, within the customer and order records used in this project, there is no observed repeat purchasing. The finding is important for customer retention analysis, but it should be treated as a characteristic of the available dataset rather than proof that these customers never purchased again outside the recorded data.

Recommendations

1. Improve Customer Retention

The business could develop follow-up campaigns after a customer's first purchase and study the products and categories associated with higher-spending customers. This is justified by Query 6's spending-tier results and Query 7's finding that all customers in the available data are classified as one-time buyers.

2. Focus on High-Performing Categories and Products

The business could review the leading health and beauty products, monitor demand, and use the results to support inventory and merchandising decisions. This recommendation is supported by Query 8's finding that it generated 9.26% of category sales value and Query 5's product-level rankings.